

Aleksandra A. Ziolkowska

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Rudolf Peierls Centre For Theoretical Physics, Beecroft Building, OX1 3PU Oxford, UK

Education

University of Oxford

Oxford, UK

DPHIL IN THEORETICAL PHYSICS. SUPERVISOR: PROF. FABIAN ESSLER

Oct. 2017 - present

- I researched open quantum systems, in particular the application of integrable methods to Lindblad dynamics.
- I worked on non-equilibrium dynamics in integrable Quantum Field Theories using numerical methods.

University of Edinburgh

Edinburgh, UK

MPhys IN PHYSICS WITH HONOURS - FIRST CLASS DEGREE, GRADUATED TOP OF THE YEAR

Sep. 2013 - Jul. 2017

- **Masters Project in Experimental Physics** “AC susceptibility measurements in heavy fermion superconductor U_6Fe ”.

Supervisor: Prof. Andrew Huxley.

I constructed an experimental apparatus and numerically analysed the data on interactions between superconductivity and charge-density waves, as well as different regimes of flux line lattice behaviour.

- **Senior Honours Project in Theoretical Physics** “Spin interactions in $ZnCr_2Se_4$ ”.

Supervisor: Dr Christopher Stock.

I constructed a mathematical model for neutron scattering experimental data; the results were published.

- 2nd year direct entry.

Nanyang Technological University

Singapore

YEAR ABROAD

Aug. 2014 - May 2015

- Theoretical and experimental training in technological applications of modern physics, such as fabrication of nanoelectronic devices, microprocessors, and photonics.

Other Research Experience

University of Edinburgh, Scottish Microelectronics Centre

Edinburgh, UK

INTERNSHIP

Jul. 2016 - Aug. 2016

- Electronics project “Polymer stabilisation of gradient-pitch liquid crystal lasers”. Supervisor: Dr Philip Hands.
- Successfully developed a novel technology of liquid crystal lasers for medical imaging, using techniques spanning the fields of physics, chemistry, electronics, and mechanical engineering.

Centre for Quantum Technologies

Singapore

INTERNSHIP

Jun. 2015 - Aug. 2015

- Experimental Physics project “Construction of a Magneto-Optical Trap for atomic laser cooling”. Supervisor: Prof. Rainer Dumke.
- Constructed a portable experimental apparatus for observing the behaviour of atoms under extreme conditions by combining laser and vacuum technologies.

Nanyang Technological University, Spintronics laboratory

Singapore

UNDERGRADUATE INTERNSHIP

Aug. 2014 - May 2015

- Experimental Physics project “Study of the domain wall injection in PMA nanowires”. Supervisor: Prof. Lew Wen Siang.
- Optimised a key process in a novel spintronics technology, which significantly lowered the power consumption of the device; used a wide range of nanofabrication techniques.

Teaching

Quantum Field Theory Tutor, University of Oxford

2019/20 and 2020/21

I taught classes to a group of 15 Master's students and graded their homework assignments. I catered to different abilities by giving extra, stimulating material to the top students and providing individual support to students that were struggling. I was invited to tutor the course again this academic year.

Hertford College Lecturer, University of Oxford

2020/21

I am teaching undergraduate courses to the physics students of the college by conducting small group tutorials tailored to the abilities of every group. As a part of the academic body I am also responsible for student welfare.

Peer Support Scheme, University of Edinburgh

2016/17

I tutored various introductory physics and mathematics courses to undergraduate students struggling with the material. The courses included Linear Algebra, Vector Calculus, Relativistic Physics and Quantum Mechanics.

Awards

Jun. 2020	Winner: Best Student Presentation	<i>Condensed Matter in All the Cities</i>
2017-2021	Engineering and Physical Sciences Research Council UK Doctoral Stipend	<i>University of Oxford</i>
2016/17	Class Medal for Physics Integrated Masters	<i>University of Edinburgh</i>
2016/17	Neil Arnott Prize for Experimental Physics	<i>University of Edinburgh</i>
2015/16	Class Medal for Physics Senior Honours	<i>University of Edinburgh</i>
2015/16	College of Science and Engineering Dean's Scholarship	<i>University of Edinburgh</i>
2015/16	The John Lang Scholarship	<i>University of Edinburgh</i>
2013/14	Year 2 Certificate of Merit for Physics	<i>University of Edinburgh</i>

Academic Conferences and Seminars

Sep. 2020	Clean and disordered systems out of equilibrium POSTER PRESENTER: "Yang Baxter Integrable Lindblad Equations"	<i>Cargèse Institute</i>
Jun. 2020	Condensed Matter in All the Cities SPEAKER: "Yang Baxter Integrable Lindblad Equations"	<i>Online</i>
Jan. 2020	Invited Seminar SPEAKER: "Yang Baxter Integrable Lindblad Equations"	<i>DIAS, Dublin</i>
Oct. 2019	Invited Seminar SPEAKER: "To Learn: Bethe Ansatz - Feynman's Last Regret"	<i>St. John's College, Oxford</i>
Apr. 2019	Student Workshop on Integrability ORGANISER	<i>University of Louvain</i>
Jun. 2017	Conference of Astronomy and Physics Students SPEAKER AND POSTER PRESENTER: "AC susceptibility measurements in heavy fermion superconductor U_6Fe "	<i>University College London</i>
Mar. 2016	EUYSRA Interdisciplinary Student Conference SPEAKER AND POSTER PRESENTER: "Study of the domain wall injection in PMA nanowires"	<i>University of Edinburgh</i>

Other Relevant Experience

Middle Common Room Committee, St. John's College *Oxford, UK*
WOMEN'S OFFICER *2018/19 and 2019/20*

I took care of the welfare of female graduate students at St. John's College and organised events promoting gender equality. I have been elected to this role twice in a row.

Conference for Undergraduate Women in Physics (CUWiP) UK *Oxford, UK*
ORGANISER *Mar. 2019*

My role in the organisation of this event involved both the recruitment of the participants as well as extending invitations to the speakers and panel members. During the event I moderated the panel discussion and took care of the smooth running of the talks.

Conference: Science Polish Perspectives, Said Business School *Oxford, UK*
SPEAKER *Nov. 2018*

I gave a popular science talk, "Nature isn't classical, dammit and why do we need quantum simulators", before an audience of 150. I engaged in discussions with entrepreneurs, activists, and policy makers from Poland interested in physics.

Marie Curious Workshop *Oxford, UK*
DEMONSTRATOR *Feb. 2018*

I volunteered in an outreach event for female secondary school students, organised to inspire them to pursue a career in STEM. I helped the participants with the workshop exercises and answered their questions about science.

Key Stage 3 Challenge Workshop *Oxford, UK*
DEMONSTRATOR *Jan. 2017*

I took part in a science outreach event aimed at primary school children from underprivileged backgrounds, organising activities and experiments on simple physics topics.

References

Prof. Fabian Essler
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Dr Olalla Castro-Alvaredo
COLLABORATOR, CITY, UNIVERSITY OF LONDON
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Prof Benjamin Doyon
COLLABORATOR, KING'S COLLEGE LONDON
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